

توثيق تثبيت Oracle Database 19c

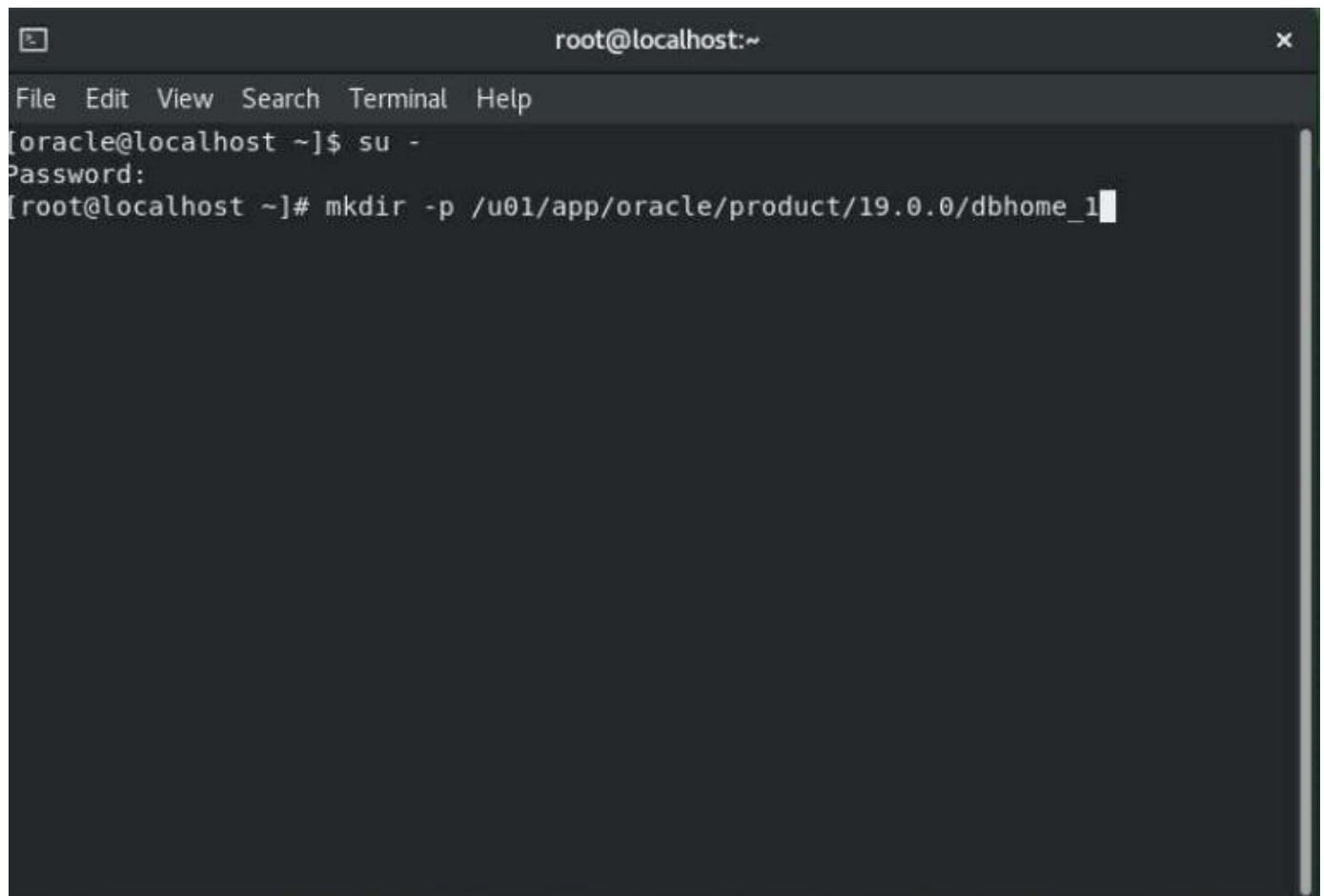
خطوات تجهيز النظام وتثبيت قاعدة بيانات Oracle

هذا المستند يوثق بالصور خطوات خطوات تجهيز النظام وتثبيت قاعدة بيانات Oracle خطوة بخطوة، كجزء من مشروع تدريب عملي ذاتي على إدارة قواعد بيانات Oracle.

عدد الخطوات الموثقة: 20

تجهيز مسار Oracle Home

تسجيل الدخول بصلاحيات root وإنشاء المسار u01/app/oracle/product/19.0.0/dbhome_1



```
root@localhost:~  
File Edit View Search Terminal Help  
[oracle@localhost ~]$ su -  
Password:  
[root@localhost ~]# mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
```

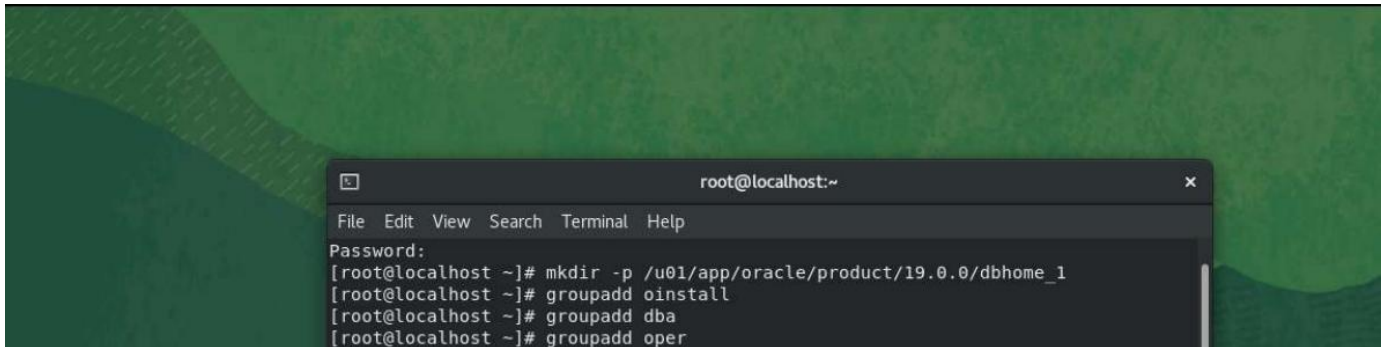
ضبط صلاحيات المستخدم oracle

تعديل المجموعة الأساسية للمستخدم oracle وإضافته لمجموعة dba، ثم التأكد من الصلاحيات الحالية.

```
[root@localhost ~]# usermod -g oinstall oracle
[root@localhost ~]# usermod -aG dba oracle
[root@localhost ~]# id
uid=0(root) gid=0(root) groups=0(root) context=unconfined_u:unconfined_r:unconfi
ned_t:s0-s0:c0.c1023
[root@localhost ~]#
```

إنشاء المجموعات المطلوبة

إنشاء مجموعات oinstall و dba و oper اللازمة لتشغيل Oracle.

A terminal window titled 'root@localhost:~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

```
Password:  
[root@localhost ~]# mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1  
[root@localhost ~]# groupadd oinstall  
[root@localhost ~]# groupadd dba  
[root@localhost ~]# groupadd oper
```

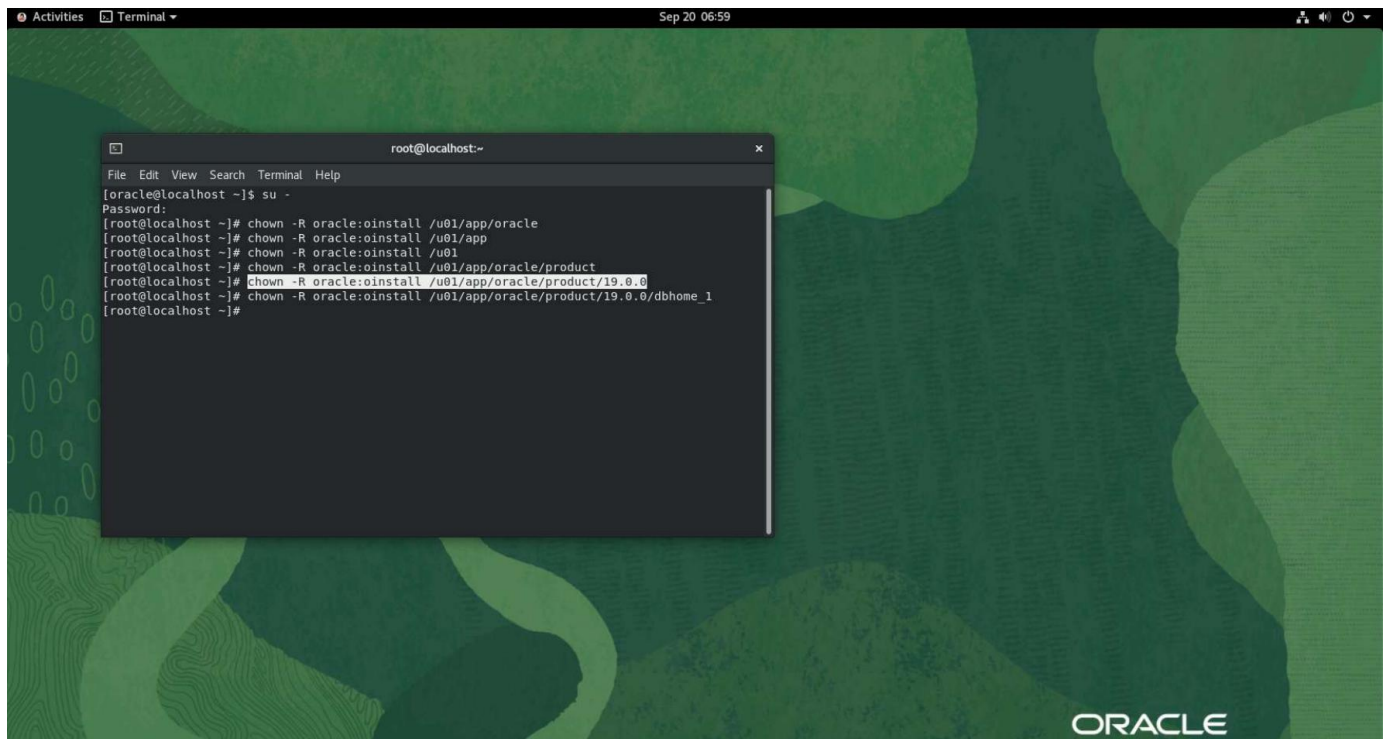
التأكد من صلاحيات oracle

تسجيل الخروج والدخول من جديد بالمستخدم oracle، والتأكد من انضمامه لمجموعتي oinstall و dba.

```
[root@localhost ~]# usermod -g oinstall oracle
[root@localhost ~]# usermod -aG dba oracle
[root@localhost ~]# id
uid=0(root) gid=0(root) groups=0(root) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[root@localhost ~]# exit
logout
[oracle@localhost ~]$ id
uid=1000(oracle) gid=1000(oracle) groups=1000(oracle),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[oracle@localhost ~]$ id oracle
uid=1000(oracle) gid=1001(oinstall) groups=1001(oinstall),10(wheel),1002(dba)
[oracle@localhost ~]$
```

الضبط الأساسي لملكية المسارات

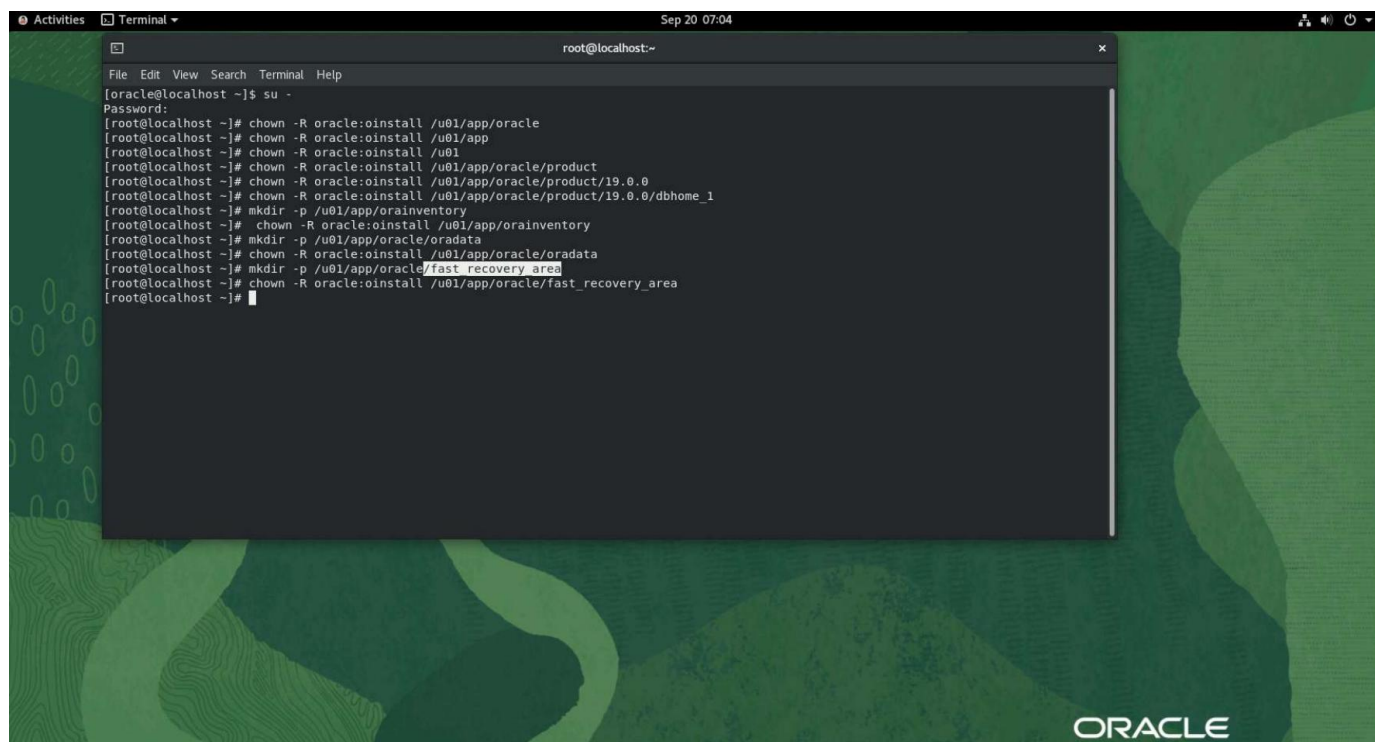
تنفيذ أوامر chown لضبط ملكية مجلدات u01/app/oracle الأساسية.



```
root@localhost:~  
File Edit View Search Terminal Help  
[oracle@localhost ~]$ su -  
Password:  
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle  
[root@localhost ~]# chown -R oracle:oinstall /u01/app  
[root@localhost ~]# chown -R oracle:oinstall /u01  
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product  
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product/19.0.0  
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product/19.0.0/dbhome_1  
[root@localhost ~]#
```

استكمال إعداد المسارات

إعادة ضبط الملكية وإنشاء مجلدات إضافية (oraInventory، oradata، fast_recovery_area) مع ضبط صلاحياتها.



```
Activities Terminal Sep 20 07:04
root@localhost:~
File Edit View Search Terminal Help
[oracle@localhost ~]$ su -
Password:
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle
[root@localhost ~]# chown -R oracle:oinstall /u01/app
[root@localhost ~]# chown -R oracle:oinstall /u01
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product/19.0.0
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/product/19.0.0/dbhome_1
[root@localhost ~]# mkdir -p /u01/app/orainventory
[root@localhost ~]# chown -R oracle:oinstall /u01/app/orainventory
[root@localhost ~]# mkdir -p /u01/app/oracle/oradata
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/oradata
[root@localhost ~]# mkdir -p /u01/app/oracle/fast_recovery_area
[root@localhost ~]# chown -R oracle:oinstall /u01/app/oracle/fast_recovery_area
[root@localhost ~]#
```

ORACLE

إعداد متغيرات البيئة وملف oraInst.loc

تحديد ORACLE_HOME و ORACLE_BASE بشكل دائم، ومحاولة إنشاء ملف oraInst.loc (كلمة المرور مموهة لأسباب أمنية).

```

Activities Terminal Sep 20 07:18
root@localhost:~
File Edit View Search Terminal Help
[oracle@localhost ~]$ export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
[oracle@localhost ~]$ export ORACLE_BASE=/u01/app/oracle
[oracle@localhost ~]$ export ORACLE_SID=orcl
[oracle@localhost ~]$ export PATH=$ORACLE_HOME/bin:$PATH
[oracle@localhost ~]$ cat > ~/.bash_profile <<'EOF'
> export ORACLE_SID=orcl
> export ORACLE_BASE=/u01/app/oracle
> export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
> export PATH=$ORACLE_HOME/bin:$PATH
> EOF
[oracle@localhost ~]$ source ~/.bash_profile
[oracle@localhost ~]$ cat > /etc/oraInst.loc <<'EOF'
> inventory_loc=/u01/app/oraInventory
> inst_group=oinstall
> EOF
bash: /etc/oraInst.loc: Permission denied
[oracle@localhost ~]$ su -
Password:
su: Authentication failure
[oracle@localhost ~]$
bash command not found...

[oracle@localhost ~]$
[oracle@localhost ~]$ su -
Password:
[root@localhost ~]# cat > /etc/oraInst.loc <<'EOF'
> inventory_loc=/u01/app/oraInventory
> inst_group=oinstall
> EOF
[root@localhost ~]# cat > /etc/oraInst.loc <<'EOF'
> inventory_loc=/u01/app/oraInventory
> inst_group=oinstall
^C
[root@localhost ~]# cat /etc/oraInst.loc
inventory_loc=/u01/app/oraInventory
inst_group=oinstall
[root@localhost ~]#

```


تثبيت أداة gcc

تثبيت المترجم gcc وتوابعه، وهو أحد المتطلبات الأساسية لتثبيت قاعدة بيانات Oracle.

```

root@localhost:~# dnf install -y gcc
Oracle Linux 8 BaseOS Latest (x86_64) 3.8 MB/s | 166 MB 00:43
Oracle Linux 8 Application Stream (x86_64) 3.1 MB/s | 86 MB 00:27
Latest Unbreakable Enterprise Kernel Release 7 for Oracle Linux 8 (x86_64) 3.5 MB/s | 120 MB 00:34
Last metadata expiration check: 0:00:47 ago on Sun 20 Sep 2026 07:09:35 AM EDT.
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
gcc x86_64 8.5.0-28.0.1.el8_10 ol8_appstream 24 M
Upgrading:
libgcc x86_64 8.5.0-28.0.1.el8_10 ol8_baseos_latest 94 k
libgomp x86_64 8.5.0-28.0.1.el8_10 ol8_baseos_latest 219 k
Installing dependencies:
cpp x86_64 8.5.0-28.0.1.el8_10 ol8_appstream 10 M
glibc-devel x86_64 2.28-251.0.2.el8 ol8_baseos_latest 89 k
isl x86_64 0.16.1-6.el8 ol8_appstream 841 k
libxcrypt-devel x86_64 4.1.1-6.el8 ol8_baseos_latest 25 k
Transaction Summary
=====
Install 5 Packages
Upgrade 2 Packages
Total download size: 35 M
Downloading Packages:
(1/7): libxcrypt-devel-4.1.1-6.el8.x86_64.rpm 58 kB/s | 25 kB 00:00
(2/7): glibc-devel-2.28-251.0.2.el8.x86_64.rpm 166 kB/s | 89 kB 00:00
(3/7): isl-0.16.1-6.el8.x86_64.rpm 986 kB/s | 841 kB 00:00
(4/7): libgcc-8.5.0-28.0.1.el8_10.x86_64.rpm 98 kB/s | 94 kB 00:00
(5/7): libgomp-8.5.0-28.0.1.el8_10.x86_64.rpm 391 kB/s | 219 kB 00:00
(6/7): cpp-8.5.0-28.0.1.el8_10.x86_64.rpm 1.6 MB/s | 10 MB 00:06
(7/7): gcc-8.5.0-28.0.1.el8_10.x86_64.rpm 2.3 MB/s | 24 MB 00:10
-----
Total 3.3 MB/s | 35 MB 00:10
Oracle Linux 8 BaseOS Latest (x86_64) 3.0 MB/s | 3.1 kB 00:00
Importing GPG key 0xAD986DA3:
Userid : "Oracle OSS group (Open Source Software group) <build@oss.oracle.com>"
Fingerprint: 76FD 3DB1 3A86 7410 B89D B10E 8256 2EA9 AD98 6DA3
From : /etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test

```

تثبيت ++gcc-c

تثبيت مترجم C++ وتوابعه لاستكمال متطلبات البناء.

```

root@localhost:~
File Edit View Search Terminal Help
root@localhost ~]# dnf install -y gcc-c++
Last metadata expiration check: 0:12:41 ago on Sun 20 Sep 2026 07:09:35 AM EDT.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
gcc-c++                                x86_64            8.5.0-28.0.1.el8_10  ol8_appstream    12 M
Upgrading:
libstdc++                              x86_64            8.5.0-28.0.1.el8_10  ol8_baseos_latest 484 k
Installing dependencies:
libstdc++-devel                         x86_64            8.5.0-28.0.1.el8_10  ol8_appstream    2.1 M
=====
Transaction Summary
=====
Install 2 Packages
Upgrade 1 Package

Total download size: 15 M
Downloading Packages:
(1/3): libstdc++-8.5.0-28.0.1.el8_10.x86_64.rpm           540 kB/s | 484 kB   00:00
(2/3): libstdc++-devel-8.5.0-28.0.1.el8_10.x86_64.rpm     1.2 MB/s | 2.1 MB   00:01
(3/3): gcc-c++-8.5.0-28.0.1.el8_10.x86_64.rpm            2.8 MB/s | 12 MB    00:04
-----
Total                                                    3.4 MB/s | 15 MB    00:04
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Upgrading : libstdc++-8.5.0-28.0.1.el8_10.x86_64 1/1
Running scriptlet: libstdc++-8.5.0-28.0.1.el8_10.x86_64 1/4
Installing : libstdc++-devel-8.5.0-28.0.1.el8_10.x86_64 2/4
Installing : gcc-c++-8.5.0-28.0.1.el8_10.x86_64 3/4
Cleanup    : libstdc++-8.5.0-21.0.1.el8.x86_64 4/4
Running scriptlet: libstdc++-8.5.0-21.0.1.el8.x86_64 4/4
Verifying  : gcc-c++-8.5.0-28.0.1.el8_10.x86_64 1/4
Verifying  : libstdc++-devel-8.5.0-28.0.1.el8_10.x86_64 2/4
Verifying  : libstdc++-8.5.0-28.0.1.el8_10.x86_64 3/4
Verifying  : libstdc++-8.5.0-21.0.1.el8.x86_64 4/4

Upgraded:
libstdc++-8.5.0-28.0.1.el8_10.x86_64
Installed:

```

libaio-devel تثبيت

تثبيت مكتبة الإدخال والإخراج غير المتزامن المطلوبة لتشغيل Oracle.

```

Activities Terminal Sep 20 07:26
root@localhost:~
File Edit View Search Terminal Help
[root@localhost ~]# dnf install -y libaio-devel
Last metadata expiration check: 0:15:17 ago on Sun 20 Sep 2026 07:09:35 AM EDT.
Dependencies resolved.
=====
Package                               Architecture Version      Repository
Size
=====
Installing:
libaio-devel                          x86_64      0.3.112-1.el8 ol8_baseos_latest
19 k
Transaction Summary
=====
Install 1 Package

Total download size: 19 k
Installed size: 8.7 k
Downloading Packages:
libaio-devel-0.3.112-1.el8.x86_64.rpm                                47 kB/s
| 19 kB 00:00
-----
Total                                47 kB/s
| 19 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : libaio-devel-0.3.112-1.el8.x86_64

```

تثبيت حزمة المتطلبات المسبقة

تثبيت oracle-database-preinstall-19c التي تضبط تلقائياً أغلب إعدادات النظام المطلوبة.

```

Sep 20 07:27
root@localhost:~
File Edit View Search Terminal Help
[root@localhost ~]# dnf install -y oracle-database-preinstall-19c
Last metadata expiration check: 0:17:31 ago on Sun 20 Sep 2026 07:09:35 AM EDT.
Dependencies resolved.
=====
Package                                Architecture      Version            Repository          Size
=====
Installing:
oracle-database-preinstall-19c         x86_64            1.0-4.el8          ol8_appstream       30 k
Installing dependencies:
ksh                                     x86_64            20120801-272.0.1.el8_10  ol8_appstream       924 k
libnsl                                  x86_64            2.28-251.0.2.el8     ol8_baseos_latest   111 k
lm_sensors-libs                        x86_64            3.4.0-23.20180522git70f7e08.el8 ol8_baseos_latest   59 k
make                                    x86_64            1:4.2.1-11.el8       ol8_baseos_latest   498 k
sysstat                               x86_64            11.7.3-13.0.3.el8_10 ol8_appstream       427 k
=====
Transaction Summary
=====
Install 6 Packages

Total download size: 2.0 M
Installed size: 6.3 M
Downloading Packages:
(1/6): lm_sensors-libs-3.4.0-23.20180522git70f7e08.el8.x86_64.rpm 143 kB/s | 59 kB 00:00
(2/6): libnsl-2.28-251.0.2.el8.x86_64.rpm 243 kB/s | 111 kB 00:00
(3/6): oracle-database-preinstall-19c-1.0-4.el8.x86_64.rpm 379 kB/s | 30 kB 00:00
(4/6): make-1:4.2.1-11.el8.x86_64.rpm 749 kB/s | 498 kB 00:00
(5/6): sysstat-11.7.3-13.0.3.el8_10.x86_64.rpm 1.3 MB/s | 427 kB 00:00
(6/6): ksh-20120801-272.0.1.el8_10.x86_64.rpm 1.8 MB/s | 924 kB 00:00
-----
Total 2.1 MB/s | 2.0 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : ksh-20120801-272.0.1.el8_10.x86_64 1/1
  Installing     : ksh-20120801-272.0.1.el8_10.x86_64 1/6
  Running scriptlet: ksh-20120801-272.0.1.el8_10.x86_64 1/6
  Installing     : make-1:4.2.1-11.el8.x86_64 2/6
  Running scriptlet: make-1:4.2.1-11.el8.x86_64 2/6
  Installing     : lm_sensors-libs-3.4.0-23.20180522git70f7e08.el8.x86_64 3/6
  Running scriptlet: lm_sensors-libs-3.4.0-23.20180522git70f7e08.el8.x86_64 3/6
  Installing     : sysstat-11.7.3-13.0.3.el8_10.x86_64 4/6
  Running scriptlet: sysstat-11.7.3-13.0.3.el8_10.x86_64 4/6

```

أدوات الواجهة الرسومية

التأكد من تثبيت أدوات xorg-x11 اللازمة لتشغيل معالج تثبيت Oracle الرسومي.

```
[root@localhost ~]# dnf install -y xorg-x11-utils
usage: dnf install [-c [config file]] [-q] [-v] [--version]
                  [--installroot [path]] [--nodocs] [--noplugins]
                  [--enableplugin [plugin]] [--disableplugin [plugin]]
                  [--releasever RELEASEVER] [--setopt SETOPTS]
                  [--skip-broken] [-h] [--allowerase] [-b | --nobest] [-C]
                  [-R [minutes]] [-d [debug level]] [--debugsolver]
                  [--showduplicates] [-e ERRORLEVEL] [--obsoletes]
                  [--rpmverbosity [debug level name]] [-y] [--assumeno]
                  [--enablerepo [repo]] [--disablerepo [repo] | --repo
[repo]] [--enable | --disable] [-x [package]]
                  [--disableexcludes [repo]] [--repofrompath [repo,path]]
                  [--noautoremove] [--nogpgcheck] [--color COLOR] [--refresh]
                  [-4] [-6] [--destdir DESTDIR] [--downloadonly]
                  [--comment COMMENT] [--bugfix] [--enhancement]
                  [--newpackage] [--security] [--advisory ADVISORY]
                  [--bz BUGZILLA] [--cve CVES]
                  [--sec-severity {Critical,Important,Moderate,Low}]
                  [--forcearch ARCH]
                  PACKAGE [PACKAGE ...]
dnf install: error: unrecognized arguments: -utils
[root@localhost ~]# dnf install -y xorg-x11-utils
Last metadata expiration check: 0:01:35 ago on Sun 20 Sep 2026 07:27:44 AM EDT.
Package xorg-x11-utils-7.5-28.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]#
```

التحقق من ملفات التثبيت والمساحة

التأكد من وجود ملفات Oracle المشتركة عبر VMWare Shared Folders وتوفير مساحة كافية على القرص.

```
[root@localhost ~]# ls -lh /mnt/hgfs/
total 0
drwxrwxrwx. 1 root root 0 Sep 19 06:42 VM-Oracle
[root@localhost ~]# ls -lh /mnt/hgfs/VM-Oracle/
total 2.9G
-rwxrwxrwx. 1 root root 2.9G Sep 12 07:10 LINUX.X64_193000_db_home.zip
drwxrwxrwx. 1 root root 4.0K Sep 19 06:45 OEM_13.5
[root@localhost ~]# df -h /u01
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/ol-root 70G  9.0G   61G  13% /
[root@localhost ~]# ls -ld /u01/app/oracle/product/19.0.0/dbhome_1
drwxr-xr-x. 2 oracle oinstall 6 Sep 20 06:44 /u01/app/oracle/product/19.0.0/dbhome_1
[root@localhost ~]#
```

تجهيز البيئة الرسومية

الانتقال لمجلد الملفات المشتركة وضبط متغير DISPLAY لتشغيل واجهة التثبيت.

```
[root@localhost ~]# ls -lh /mnt/hgfs/
total 0
drwxrwxrwx. 1 root root 0 Sep 19 06:42 VM-Oracle
[root@localhost ~]# ls -lh /mnt/hgfs/VM-Oracle/
total 2.9G
-rwxrwxrwx. 1 root root 2.9G Sep 12 07:10 LINUX.X64_193000_db_home.zip
drwxrwxrwx. 1 root root 4.0K Sep 19 06:45 OEM 13.5
[root@localhost ~]# df -h /u01
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/ol-root 70G  9.0G   61G  13% /
[root@localhost ~]# ls -ld /u01/app/oracle/product/19.0.0/dbhome_1
drwxr-xr-x. 2 oracle oinstall 6 Sep 20 06:44 /u01/app/oracle/product/19.0.0/dbhome_1
[root@localhost ~]# exit
logout
[oracle@localhost ~]$ cd /mnt/hgfs/
[oracle@localhost hgfs]$ ls
VM-Oracle
[oracle@localhost hgfs]$ cd VM-Oracle
[oracle@localhost VM-Oracle]$ ls
LINUX.X64_193000_db_home.zip  OEM 13.5
[oracle@localhost VM-Oracle]$ export display = :0
bash: export: `=': not a valid identifier
bash: export: `:0': not a valid identifier
[oracle@localhost VM-Oracle]$ export DISPLAY=:0
[oracle@localhost VM-Oracle]$
```

فك ضغط ملفات Oracle Database

استخراج محتويات ملف LINUX.X64_193000_db_home.zip داخل مجلد Oracle Home.

```
[oracle@localhost VM-Oracle]$ unzip /mnt/hgfs/VM-Oracle/LINUX.X64_193000_db_home.zip \
> -d /u01/app/oracle/product/19.0.0/dbhome_1
Archive: /mnt/hgfs/VM-Oracle/LINUX.X64_193000_db_home.zip
  creating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/
  creating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/admin/
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/admin/drdsqtt_translator_setup.sql
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/admin/drdaas.ora
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/admin/drdaaslib.sql
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/admin/drdaaslib.sql
  creating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/lib/
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/lib/s0dpspmain.o
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/lib/s0dpsscmain.o
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/drdaas/lib/s0dpssmain.o
  creating: /u01/app/oracle/product/19.0.0/dbhome_1/instantclient/
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/instantclient/libsqlplusic.so
 inflating: /u01/app/oracle/product/19.0.0/dbhome_1/schagent.conf
  creating: /u01/app/oracle/product/19.0.0/dbhome_1/opmn/
```


تشغيل معالج التثبيت

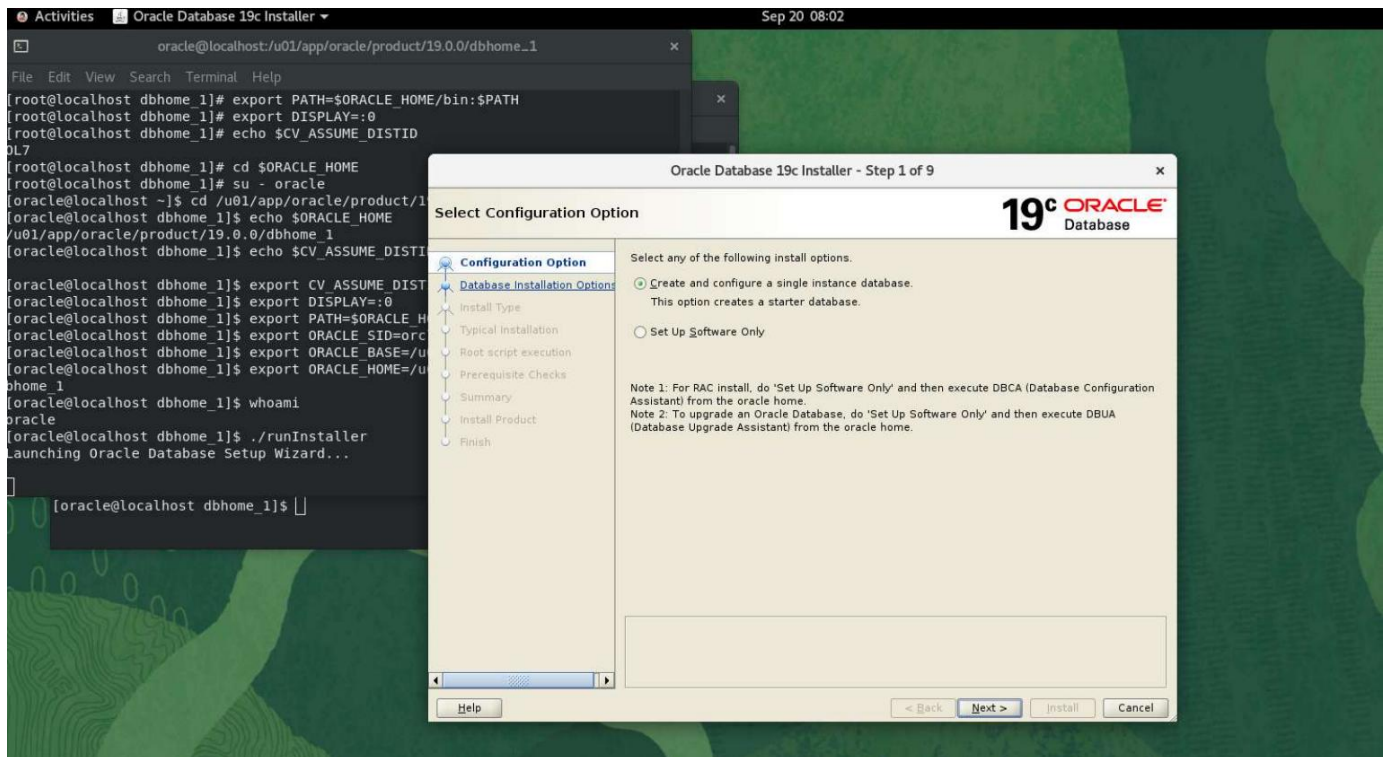
الدخول لمجلد Oracle Home والتأكد من ملفاته، ثم تشغيل الأمر runInstaller لبدء معالج التثبيت الرسومي.

```

oracle@localhost: /u01/app/oracle/product/19.0.0/dbhome_1
File Edit View Search Terminal Help
[oracle@localhost ~]$ cd /u01/app/oracle/product/19.0.0/dbhome_1
[oracle@localhost dbhome_1]$ ls
addnode      deinstall    javavm      OPatch      R            sqlj
apex          demo         jdbc        opmn        racg         sqlpatch
assistants   diagnostics  jdk         oracore     rdbms       sqlplus
bin           dmdu         jlib        ord         relnotes    srvnm
clone        drdaas       ldap        ords        root.sh      suptools
crs           dv           lib         oss         root.sh.old  ucp
css           env.ora      md          oui         root.sh.old.1 usm
ctx           has          mgw         owm         runInstaller utl
cv            hs           network     perl        schagent.conf wwg
data          install      nls         plssql      sdk          xdk
dbjava        instantclient odbc        precomp    slax
dbs           inventory    olap        OPatch     sqldeveloper
[oracle@localhost dbhome_1]$ echo $DISPLAY
:0
[oracle@localhost dbhome_1]$ ./runInstaller
  
```

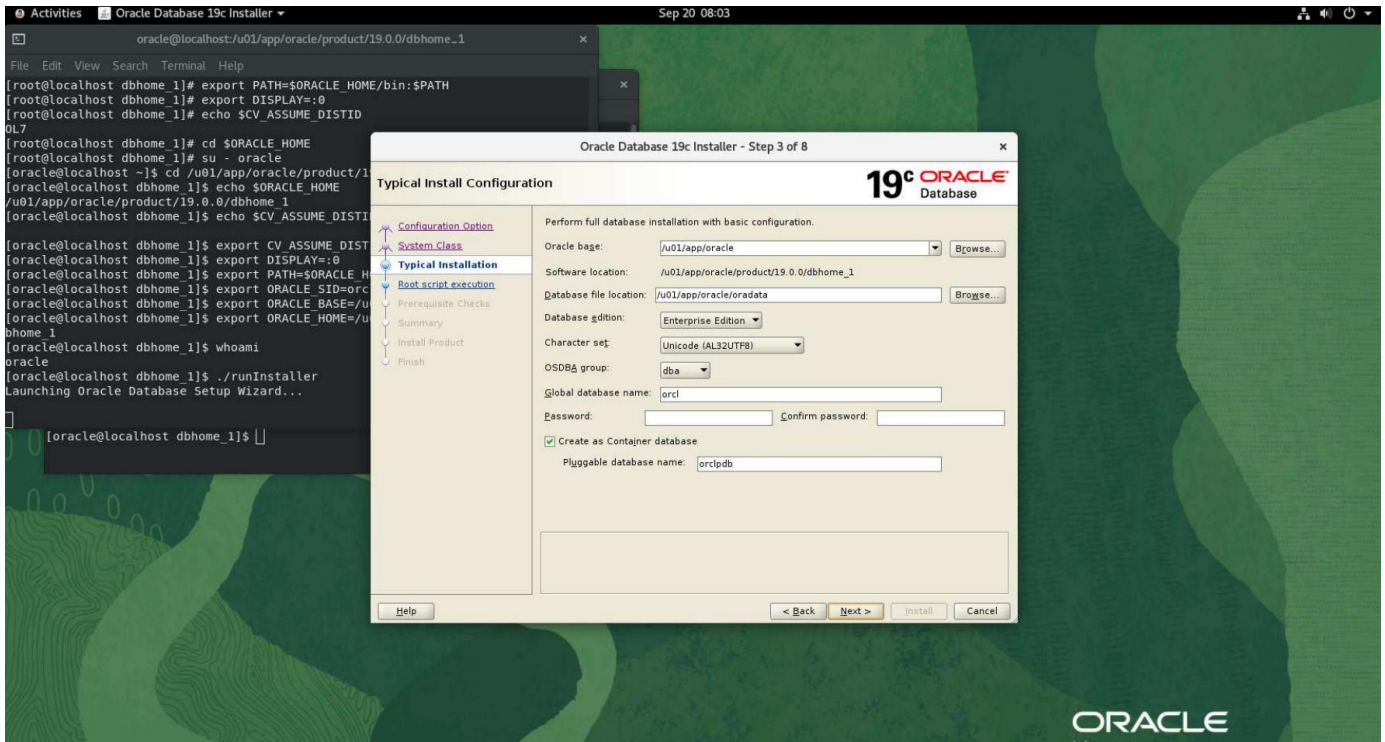
اختيار نوع التثبيت

تحديد خيار إنشاء وتهيئة قاعدة بيانات (Single Instance) من معالج Oracle 19c.



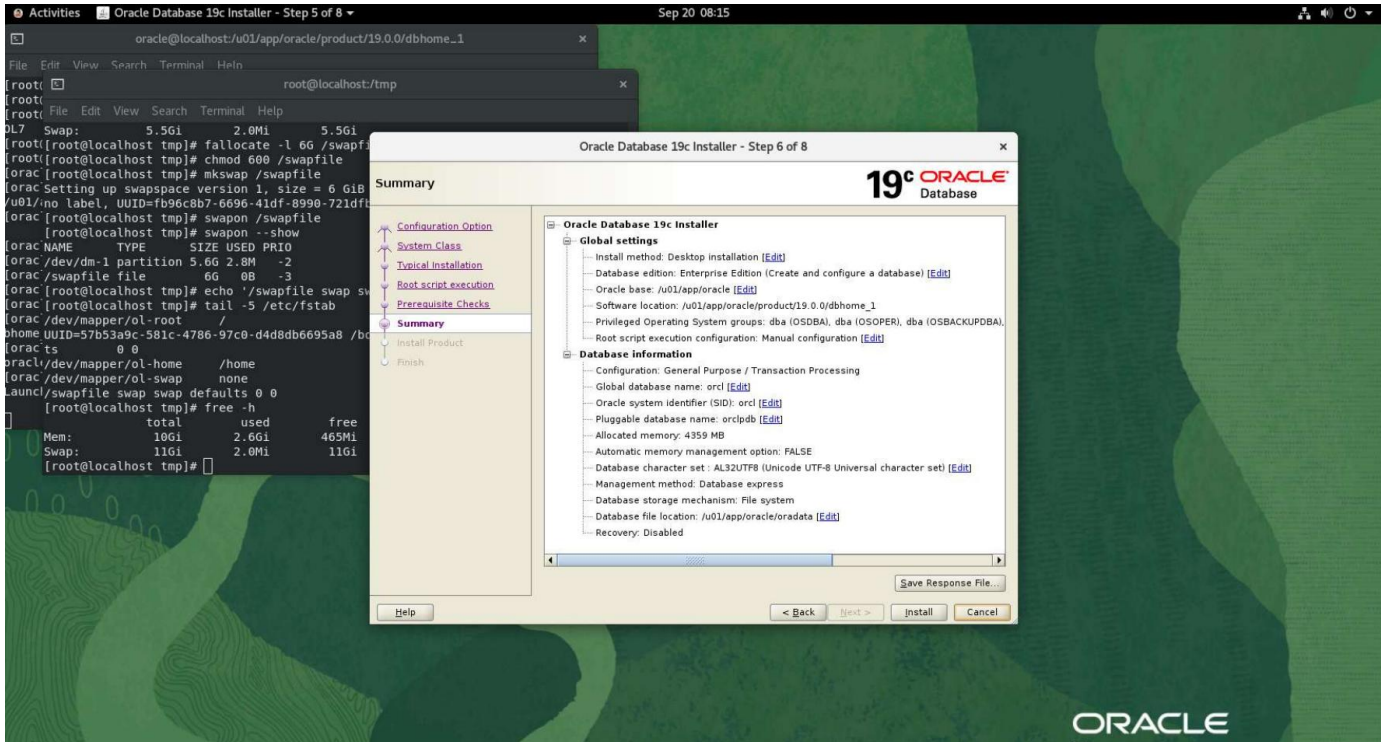
الإعدادات الأساسية لقاعدة البيانات

تحديد مسارات Oracle Base وقاعدة البيانات، واسم قاعدة البيانات orcl ونوعها كـ Container Database.



مراجعة الإعدادات وضبط الذاكرة الافتراضية

مراجعة ملخص الإعدادات النهائي، مع إعداد ملف swap إضافي لتلبية متطلبات الذاكرة.



اكتمال تثبيت قاعدة البيانات

ظهور رسالة نجاح التثبيت ورابط الوصول لـ Oracle Enterprise Manager Express.

